

impact decays. Investors must understand differences in dissipation rate when they structure trade schedules otherwise they may pay prices much higher than expected.

Figure 4.3c illustrates prices that could occur under a fast and a slow decay rate under the same sequence of trades. In this example, the starting price is \$30.00 and each trade’s market impact is \$0.25, with \$0.20 temporary and \$0.05 permanent. The first trade price under both trajectories is \$30.25. The second trade price is comprised of temporary and permanent impact for the second trade, plus permanent impact of the first trade plus the remaining temporary impact of the first trade. A fast decay rate will cause investors to incur a lower amount of temporary impact than a slow decay rate. In the example, the second trade price is \$30.30 for the fast decay rate and \$30.38 for the slow decay rate. The slower decay rate causes the investor to incur \$0.08/share more than the fast decay rate due to the amount of temporary impact still present in the market from the first trade. The spacing between these trades was just long enough for the temporary impact to dissipate fully for the fast decay function but not nearly enough for the slow decay function. The third trade price is equal to the permanent impact from the first two trades, plus the temporary impact amount of the first trade still, plus the temporary impact of the second trade price (if temporary impact still exists at the time of these trades). Under the fast decay schedule the third trade price is \$30.35, which is comprised of permanent impact from all three trades (\$0.05/share each) and temporary impact from the third trade only (\$0.20/share) since the temporary impact from the preceding two trades has already been fully dissipated in the market. Under the slow decay rate the third trade price is \$30.46, which is comprised of permanent impact from all three trades (0.05/share each—permanent impact is a cumulative effect) plus \$0.033/share of remaining temporary impact from the first trade, plus \$0.081/share of remaining temporary impact from the second trade, plus \$0.20/share of temporary impact from the third trade. We see that the average execution price is \$30.30 for the fast decay rate and \$30.37 for the slow decay rate. Investors must understand the temporary impact function’s decay rate to avoid the cumulative effect of temporary impact. The prices under the different decay schedules are shown in Table 4.1.

Table 4.1 Temporary Impact Rate of Decay				
Temporary Impact	Trade 1	Trade 2	Trade 3	Avg Price
Fast Decay	\$30.25	\$30.30	\$30.35	\$30.30
Slow Decay	\$30.25	\$30.38	\$30.46	\$30.37